HarmonyForge

A deterministic co-creative system for symbolic music arrangement

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LATE-BREAKING DEMO

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DEMO SOURCE

1 WHY The parts do not match the room.

2 DRAFT A solver writes SATB by the rules.

3 EDIT You move the notes by hand.

4 ASK A tutor explains. You confirm.

Story spine · Hard rules draft → your hands edit → theory explains on demand → you stay the author.

01 — THE STALL

The room is ready. The parts are not.

Ensembles stall when the score lacks a line for who showed up. Generative models draft fast — then leave voice-leading and range errors for musicians to fix.

HarmonyForge's answer: draft under hard rules, edit by hand, explain on demand.

N = 21
Formative interviews with musicians

Clef labor
Hours of hand transposition before anyone can play.

Sitting out
A player waits — no written line fits their instrument.

02 — LOGIC CORE

A search, not a sample.

Infers chords from your melody and writes SATB under Aldwell & Schachter hard barriers — then maps the scaffold onto whoever showed up (even flute + trombone + violin).

1 **Deterministic solver**
Greedy-then-backtracking. No sampled tokens.

2 **Common-practice SATB**
Four voices in range, from the melody you upload.

03 — TACTILE SANDBOX

Your hands on the staff.

After generation the score is yours. Edit on a patched RiffScore canvas — note-level collaging, not another prompt. Export PDF, MusicXML, WAV, or MIDI when you are done.

A **Direct manipulation**
Click a note. Drag the pitch. Hear it now.

B **Keep or rewrite**
Regenerate a measure — or place every pitch.

04 — THEORY INSPECTOR

Closed until you ask.

Click a note for deterministic interval and rule checks. Chat is retrieval-augmented from *Open Music Theory*. Suggests repairs — never writes without your confirm.

1 **Facts first**
The checker is code. The tutor waits until you open it.

2 **Grounded, not invented**
Explanations from a curated theory lexicon.

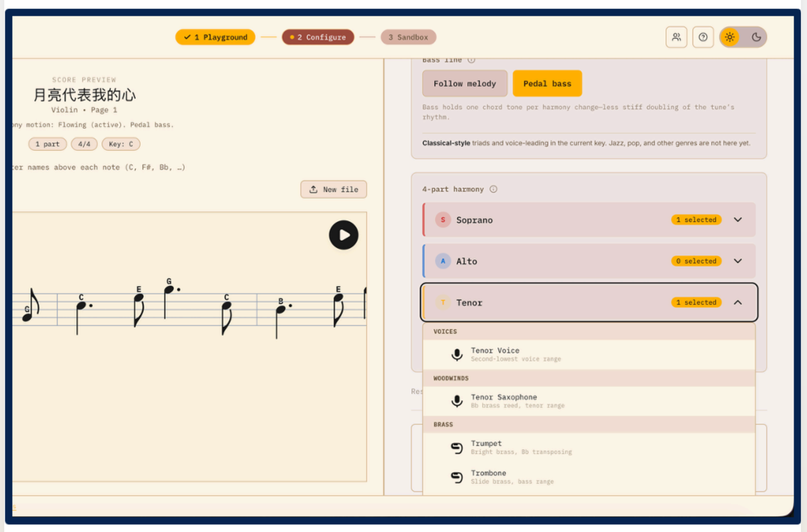


Fig. 1 · Logic Core — SATB scaffold remapped to your ensemble.

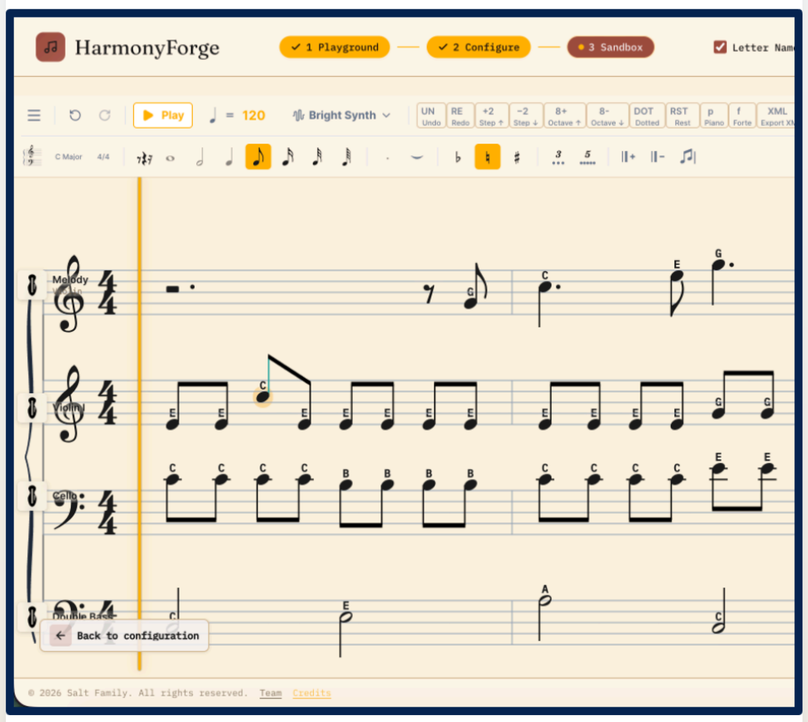


Fig. 2 · Sandbox — melody plus parts. You move the pitch.

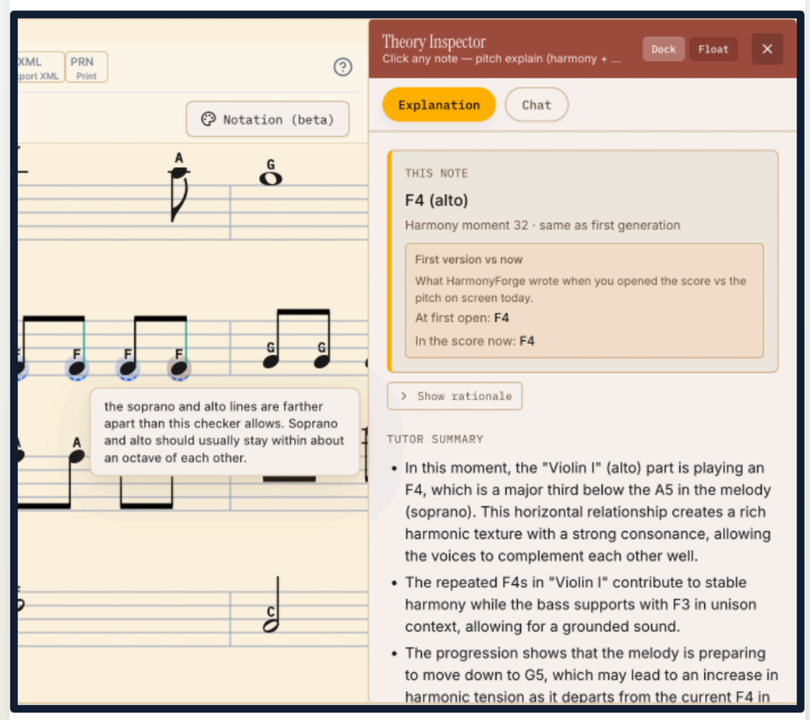


Fig. 3 · Inspector — facts + a suggested repair. You decide.

PROOF — HARD BARRIERS

What the search cannot cross

Range
Each SATB voice stays in compass.

Spacing
S–A, A–T ≤ 8ve; T–B ≤ 12th.

Crossing
No crossing: S ≥ A ≥ T ≥ B.

Parallels
No parallel fifths or octaves.

Motion heuristic
Prefer small pitch steps (Fux-inspired) — a preference, not a wall.

PROOF — VS. LLM-ONLY

Why not an LLM-only arranger?

QUESTION	TYPICAL GENERATOR	HARMONYFORGE
How it drafts	Probabilistic tokens	**Deterministic SATB search**
Illegal motion	Repair after the fact	**Hard barriers cannot be crossed**
Explanation	Opaque structure	**Fact block + Open Music Theory**
Who writes	The model	**You confirm every write**

05 — AT THE BOOTH

Five minutes. This room is the ensemble.

1 **Upload & configure** MusicXML, MIDI, or PDF. Odd mix: flute, trombone, violin.

2 **Generate** Logic Core writes playable, rule-checked parts.

3 **Break a rule** Move a note. Make parallel octaves on purpose.

4 **Inspect** Read red/blue facts. Ask the Tutor how to fix them.

5 **Export** Letter names on. Take PDF, MusicXML, WAV, or MIDI.

TAKEAWAY

Keep the human as author. Automate the mechanical work — not the musical decision.

CONTRIBUTIONS

- Deterministic SATB Logic Core with hard barriers
- Note-level sandbox that preserves authorship
- On-demand Inspector that explains, never drafts

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